

The algebra $\mathcal{U}$

We let $\mathcal{U}$ be the quantum group associated to a generalized Cartan matrix $C=(c_{ij})$.

Lemma. There is an algebra homomorphism $\chi: f \rightarrow \hat{\mathcal{U}}$ given by $\Theta_i \mapsto E_i' = (v_i - v_i^{-1}) K_i^{-1} E_i$.

Let $\mathcal{U}^+$ be the image of χ and $\mathcal{U}^0$ be the subalgebra generated by $K_i^{\pm 1}$, $i \in \mathbb{I}$. We note that $\mathcal{U}^0 \otimes \mathcal{U}^0 \otimes \mathcal{U}^+ \rightarrow \mathcal{U}$ and $\mathcal{U}^+ \otimes \mathcal{U}^0 \otimes \mathcal{U} \rightarrow \mathcal{U}$ have the same image in $\mathcal{U}$.

Denote this image by $\mathcal{U}'$. In fact, E_i' and F_j satisfy the following relation:

$$E_i' F_j = v^{(\alpha_i, \alpha_j)} F_j E_i' + \delta_{ij} (1 - K_i^{-2})$$

Let $\mathcal{U}$ be the quotient of $\mathcal{U}'$ by the ideal generated by $K_i^{\pm 1}$. Then it is clear that

(a) $\mathcal{U}^0 \otimes \mathcal{U}^+ \rightarrow \mathcal{U}$, $\mathcal{U}^+ \otimes \mathcal{U} \rightarrow \mathcal{U}$ are iso of vector spaces.

(b) $\mathcal{U}$ is defined by the generators E_i, ϕ_i , $i \in \mathbb{I}$ and the relations $E_i \phi_j = v^{(\alpha_i, \alpha_j)} \phi_j E_i + \delta_{ij}$ for all $i, j \in \mathbb{I}$, together with some relations for E_i and ϕ_i . Here E_i, ϕ_i are the images of E_i' and F_i .

Now let $r: f \rightarrow f$ be the $\mathbb{Q}(v)$ -linear map defined by

$$r(w) = 0, \quad r(\Theta_j) = \delta_{ij}, \quad r(xy) = r(w)y + v^{(\alpha_i, \alpha_w)} x r(y)$$

for x, y homogeneous. By induction one easily see that $r(x) = \Theta_i \otimes r(x) + \dots$

Let $(,)$ be the bilinear pairing on f defined by

$$(\Theta_i, \Theta_j) = \frac{\delta_{ij}}{1 - v_i^{-2}}, \quad (x, y, y_1) = (r(w, y, \Theta y_1), \quad (x, \pi_1, y) = (x, \Theta x, r(y))$$

Then we see that

$$(\Theta_i, y, x) = (\Theta_i, \Theta y, r(x)) = (\Theta_i, \Theta_i)(y, r(w)).$$

Lemma 1. For $i \in \mathbb{I}$, we define $\phi_i: f \rightarrow f$ to be left multiplication by Θ_i and $\varepsilon_i: f \rightarrow f$ to be r .

(a) ϕ_i, ε_i make f into a left $\mathcal{U}$ -module.

(b) $\varepsilon_i: f \rightarrow f$ is locally nilpotent for $i \in \mathbb{I}$.

Proof. Note that for $y \in f$ homogeneous.

$$\varepsilon_i \phi_j(y) = r(\Theta_j y) = \delta_{ij} \cdot y + v^{(\alpha_i, \alpha_j)} \Theta_j \cdot r(y) = v^{(\alpha_i, \alpha_j)} \phi_j \varepsilon_i(y) + \delta_{ij} \cdot y$$

Now let $F = F(0)$ be a homogeneous element in the defining ideal I of f . Then from definition we have $F(\phi_i) = 0$ on f . Let $F' = \sigma(F) \in I$. Then from the identity

$$(x, \varepsilon_i(y)) = (1 - \bar{v}_i^2)(\phi_i(x, y))$$

we see that $(x, F(\varepsilon_i)y) = \lambda \cdot (F'(\phi_i)x, y) = 0$. Since this is true for all x, y homogeneous, we conclude that $F(\varepsilon_i) = 0$.

Finally, if $x \in f_v$ then $\sigma(x) \in f_{v-\alpha_i}$ or $\sigma(x) = 0$. Therefore ε_i is locally nilpotent.

Lemma 2. There is a unique algebra homomorphism $d: \mathcal{U} \rightarrow \mathcal{U} \otimes \mathcal{U}$ such that

$$d(\phi_i) = \phi_i \otimes K_i^{-1} + 1 \otimes F_i, \quad d(\varepsilon_i) = \varepsilon_i \otimes K_i^{-1} + (v_i - \bar{v}_i) 1 \otimes K_i^{-1} \varepsilon_i$$

Proof. Note that the coproduct $\Delta: \mathcal{U} \rightarrow \mathcal{U} \otimes \mathcal{U}$ satisfies

$$\Delta(F_i) = F_i \otimes K_i^{-1} + 1 \otimes F_i.$$

$$\begin{aligned} \Delta(\varepsilon'_i) &= \Delta((v_i - \bar{v}_i) K_i^{-1} \varepsilon_i) = (v_i - \bar{v}_i) K_i^{-1} \varepsilon_i \otimes K_i^{-1} + (v_i - \bar{v}_i) 1 \otimes K_i^{-1} \varepsilon_i \\ &= \varepsilon'_i \otimes K_i^{-1} + (v_i - \bar{v}_i) 1 \otimes K_i^{-1} \varepsilon_i \end{aligned}$$

Hence Δ restricts to an algebra homomorphism $\Delta: \mathcal{U}' \rightarrow \mathcal{U}' \otimes \mathcal{U}$, and induces an algebra homomorphism d .

Kashimura operators in rank 1.

• Case of category $\mathcal{D}_i$.

Define $\mathcal{D}_i$ to be the category whose objects are $\mathbb{Q}(v)$ -vector spaces P provided with $\mathbb{Q}(v)$ -linear maps $\varepsilon_i, \phi_i: P \rightarrow P$ such that ε_i is locally nilpotent and

$$\varepsilon_i \phi_i = v_i^2 \phi_i \varepsilon_i + 1 \quad (*)$$

The morphisms in $\mathcal{D}_i$ are $\mathbb{Q}(v)$ -linear maps commuting with ε_i, ϕ_i .

For $P \in \mathcal{D}_i$ and $s \in \mathbb{Z}$, let $\phi_i^{(s)}: P \rightarrow P$ be defined by $\phi_i^s / [s]_i!$ if $s \geq 0$ and 0 if $s < 0$.

By induction on N we find that

$$\varepsilon_i \phi_i^{(N)} = v_i^{2N} \phi_i^{(N)} \varepsilon_i + v_i^{N-1} \phi_i^{(N-1)} \quad \text{for all } N \in \mathbb{Z}. \quad (**)$$

For $t \geq 0$, we define the operators

$$\Pi_t := \sum_{s \geq 0} (-1)^s v_i^{\frac{s(s-1)}{2}} \phi_i^{(s)} \varepsilon_i^{s+t} : P \rightarrow P$$

Since ε_i is locally finite, this is well-defined. For $N \geq 0$ we set

$$P^{(0)} := \{x \in P \mid \varepsilon_i(x) = 0\}, \quad P^{(N)} := \phi_i^{(N)} P^{(0)}.$$

Lemma 3. (a) We have $\varepsilon_i \Pi_t = 0$ for all $t \geq 0$.

(b) We have $\sum_{t \geq 0} v_i^{-\frac{t(t-1)}{2}} \phi_i^{(t)} \Pi_t = 1$. This sum is well-defined since for any $x \in P$, we have $\Pi_t(x) = 0$ for large t .

(c) We have a direct sum decomposition $P = \bigoplus_{N \geq 0} P^{(N)}$ as a $\mathbb{Q}(v)$ -vector space. Moreover, for $N \geq 0$ the map $\phi_i^{(N)}$ restricts to an isomorphism $P^{(0)} \xrightarrow{\sim} P^{(N)}$.

(d) $\phi_i : P \rightarrow P$ is injective.

Proof. By (a*) we have

$$\begin{aligned} \varepsilon_i \Pi_t &= \sum_{s \geq 0} (-1)^s v_i^{\frac{s(s-1)}{2}} (v_i^{2s} \phi_i^{(s)} \varepsilon_i^{s+t+1} + v_i^{s-1} \phi_i^{(s-1)} \varepsilon_i^{s+t}) \\ &= \sum_{s \geq 0} (-1)^s \phi_i^{(s)} \varepsilon_i^{s+t+1} \left(v_i^{\frac{s(s-1)}{2} + 2s} - v_i^{\frac{s(s-1)}{2} + s} \right) \\ &= 0. \end{aligned}$$

This proves (a).

To prove (b) note that

$$\begin{aligned} \sum_{t \geq 0} v_i^{-\frac{t(t-1)}{2}} \phi_i^{(t)} \Pi_t &= \sum_{t \geq 0} \sum_{s \geq 0} (-1)^s v_i^{-\frac{t(t-1)}{2} + \frac{s(s-1)}{2}} \phi_i^{(t)} \phi_i^{(s)} \varepsilon_i^{s+t} \\ &= \sum_{t, s \geq 0} (-1)^s v_i^{\frac{s(s-1) - t(t-1)}{2}} \begin{bmatrix} t+s \\ t \end{bmatrix}_i \phi_i^{(t+s)} \varepsilon_i^{t+s} \\ &= \sum_{t \geq s} (-1)^s v_i^{\frac{1}{2}(s^2 - (t-1)(t-s-1))} \begin{bmatrix} t \\ s \end{bmatrix}_i \phi_i^{(t)} \varepsilon_i^{(t)} \\ &= \sum_{t \geq s} (-1)^s v_i^{\frac{1}{2}(t-t^2)} \cdot v_i^{\frac{(t-1)s}{2}} \begin{bmatrix} t \\ s \end{bmatrix}_i \phi_i^{(t)} \varepsilon_i^{(t)} \\ &= 1 \end{aligned}$$

where we used $\sum (-1)^s v_i^{\frac{s(s-1)}{2}} \begin{bmatrix} t \\ s \end{bmatrix}_i = 0$ for $t > 0$.

We now prove (c). If $x \in P$ then by (b) we have

$$x = \sum_N \phi_i^{(N)}(x_N)$$

where $x_N = v_i^{-\frac{N(N-1)}{2}} \Pi_N(x)$. By (a) we have $\Pi_N(x) \in P(\omega)$, so $P = \sum_N P(N)$. It remains to show that each x_N is unique. It suffices to prove the following:

If $0 = \sum_{N \geq 0} \phi_i^{(N)}(x_N)$, where $x_N \in P(\omega)$ and $x_N = 0$ for $N > N_0$, then $x_{N_0} = 0$.

We prove this by induction on N_0 . If $N_0 = 0$ then this is clear. Now assume $N_0 \geq 1$ and apply ε_i to $\sum_N \phi_i^{(N)}(x_N)$ and use (20), we get $0 = \sum_{N \geq 0} v_i^{N-1} \phi_i^{(N-1)}(x_N)$. Using the induction hypothesis we then get $x_{N_0} = 0$. Finally, (d) follows from (c).

We define linear maps $\tilde{\phi}_i, \tilde{\varepsilon}_i: P \rightarrow P$ by

$$\tilde{\phi}_i(\phi_i^{(N)} y) = \phi_i^{(N+1)} y, \quad \tilde{\varepsilon}_i(\phi_i^{(N)} y) = \phi_i^{(N-1)} y \quad \text{for all } y \in P(\omega).$$

A symmetric bilinear form $(\cdot, \cdot): P \times P \rightarrow \mathbb{Q}(\omega)$ is called admissible if

$$(x, \varepsilon_i(y)) = (1 - v_i^{-2})(\phi_i(x), y) \quad \text{for all } x, y \in P.$$

Besides $A = \mathbb{Z}[v, v^{-1}]$, we also need the following subrings of $\mathbb{Q}(\omega)$:

$$A = \mathbb{Q}[\mathbb{Z}[v^{-1}]] \cap \mathbb{Q}(\omega), \quad A_2 := \mathbb{Z}[\mathbb{Z}[v^{-1}]] \cap \mathbb{Q}(v).$$

Let B be a basis of the $\mathbb{Q}(\omega)$ -vector space P . We say that B is integral if the A -submodule ${}_A P$ of P generated by B is stable under $\varepsilon_i, \phi_i^{(t)}: P \rightarrow P$ for all $t \geq 0$.

Assume that we are given an admissible form $(\cdot, \cdot)$ and an integral basis B of P which is almost orthonormal (i.e. $(b, b) \in \delta_{b,b} + v^{-1}A_2$). Define

$$\mathbb{L}(P) := \{x \in {}_A P \mid (x, x) \in A\}.$$

Lemma 4. (a) $\mathbb{L}(P)$ is a $\mathbb{Z}[v^{-1}]$ -submodule of ${}_A P$ and B is a basis of it.

(b) Let $x \in {}_A P$ be such that $(x, x) \in 1 + v^{-1}A$. Then there exists $b \in B$ such that $x \equiv \pm b \pmod{v^{-1}\mathbb{L}(P)}$.

(c) Let $x \in {}_A P$ be such that $(x, x) \in v^{-1}A$. Then $x \in v^{-1}\mathbb{L}(P)$.

Proof. Let $x \in P$. Assume that $x \neq 0$. Then we can write uniquely $x = \sum_{b \in B} c_b b$ with $c_b \in \mathbb{Q}(v)$. Since only finitely many c_b are nonzero, we can find uniquely $t \in \mathbb{Z}$ and $p_b \in \mathbb{Q}$ such that for all b we have $v^{-t} c_b - p_b \in v^{-1}A$. Then we have

$$(x, x) = \sum_b p_b^2 v^{2t} \pmod{v^{2t-1}A}.$$

Note that $\sum_b p_b^2$ is a positive rational number, so $(x, x) \in A$ iff $t \geq 0$. This is equivalent to the condition that $c_b \in A$ for all b , so (a) follows.

Now if $(x, x) \in v^{-1}A$ then we must have $t < 0$, so $c_b \in v^{-1}A$ for all b and (a) follows. If $x \in \Delta P$ and $(x, x) \in 1 + v^{-1}A$, then we must have $t = 0$ and $\sum_b p_b^2 = 1$ with $p_b \in \mathbb{Z}$ (since $x \in \Delta P$).

Then $p_b = \pm 1$ for a unique b , and $p_b = 0$, so (b) follows.

Lemma 5. Let $y \in \Delta P$ be such that $\varepsilon_i y = 0$, and $n \geq 0$. Then we have $(\phi_i^{(n)} y, \phi_i^{(n)} y) = \pi_n (y, y)$,

where $\pi_n \in 1 + v^{-1}\mathbb{Z}[v^{-1}]$.

Proof. Using induction, it suffices to prove

$$(\phi_i^{(n+1)} y, \phi_i^{(n+1)} y) = \pi_n (\phi_i^{(n)} y, \phi_i^{(n)} y)$$

where $\pi_n \in 1 + v^{-1}\mathbb{Z}[v^{-1}]$. Now we have

$$\begin{aligned} (\phi_i^{(n+1)} y, \phi_i^{(n+1)} y) &= (\varepsilon_{n+1}^{-1} \phi_i^{(n)} y, \phi_i^{(n+1)} y) \\ &= (1 - v_i^{-2})^{-1} \varepsilon_{n+1}^{-1} \cdot (\phi_i^{(n)} y, \varepsilon_i \phi_i^{(n+1)} y) \\ &= (1 - v_i^{-2})^{-1} \varepsilon_{n+1}^{-1} \cdot v_i^2 (\phi_i^{(n)} y, \phi_i^{(n)} y) \end{aligned}$$

It then suffices to note that $v_i^2 \varepsilon_{n+1}^{-1} \in 1 + v^{-1}\mathbb{Z}[v^{-1}]$.

Lemma 6. (a) Let $x \in \Delta P$ and write $x = \sum_{N \geq 0} \phi_i^{(N)} x_N$, where $x_N \in P(v)$ are zero for large N .

Then $x_N \in \Delta P$ for all N .

(b) If $x \in \mathbb{I}(P)$ then each x_N and $y_N := \phi_i^{(N)}(x_N)$ is in $\mathbb{I}(P)$. If in addition

$(x, x) \in 1 + v^{-1}A$, then there exists $N_0 \geq 0$ and $b \in B$ such that $x_{N_0} = \pm b \pmod{v^{-1}\mathbb{I}(P)}$ and $x_N, y_N \in v^{-1}\mathbb{I}(P)$ for $N \neq N_0$.

Proof. We prove (a). We have $x_N = v_i^{-\frac{N(N+1)}{2}} \pi_N(x)$. Since ΔP is stable under ε_i and $\phi_i^{(N)}$, we see that ΔP is stable under π_N . Hence $x_N \in \Delta P$.

Next we show that the subspaces $\phi_i^{(N)} P(i)$, $\phi_i^{(N')} P(i)$ are orthogonal to each other under (.) if $N \neq N'$. We argue this by induction on $N+N'$. If $N \geq 1$ we have for $z, z' \in P(i)$ that

$$(\phi_i^{(N)} z, \phi_i^{(N')} z') = \lambda \cdot (\phi_i^{(N-1)} z, \phi_i^{(N'-1)} z')$$

for some scalar λ , so it is zero by induction. If $N' \geq 1$ we can prove similarly. If $N \leq 0$ or $N' \leq 0$ the claim is trivial.

Now let $x \in {}_\lambda P$ be nonzero. We have $(x, x) = \sum_N (y_N, y_N)$. We can find $t \in \mathbb{Z}$ s.t. $v^{-t} y_N \in \mathbb{I}(P)$ for all N , and $v^{-t+1} y_N \notin \mathbb{I}(P)$ for some N . Then there exist integers $a_N \geq 0$, not all zero, such that $v^{-2t} (y_N, y_N) - a_N \in v^{-1} A$ for all N . Hence

$$(c) \quad v^{-2t} (x, x) - \sum_N a_N \in v^{-1} A \quad \text{and} \quad \sum_N a_N > 0$$

If $x \in \mathbb{I}(P)$ then (c) shows that $t \leq 0$, hence $y_N \in \mathbb{I}(P)$ for all N . Using the previous lemma we see that $x_N \in \mathbb{I}(P)$ for all N . If in addition we have $(x, x) \in 1 + v^{-1} A$, then (c) shows $t=0$ and $a_{N_0}=1$ for some N_0 and $a_N=0$ for all $N \neq N_0$. In other words,

$$(y_{N_0}, y_{N_0}) \in 1 + v^{-1} A, \quad (y_N, y_N) \in v^{-1} A, \quad \forall N \neq N_0.$$

Using the previous lemma we get $(x_{N_0}, x_{N_0}) \in 1 + v^{-1} A$ and $(x_N, x_N) \in v^{-1} A$ for $N \neq N_0$.

The second assertion then follows.

Lemma 7. The maps $\hat{\phi}_i, \hat{\xi}_i: P \rightarrow P$ map $\mathbb{I}(P)$ into itself.

Proof. Let $x \in \mathbb{I}(P)$. We must show that $\hat{\phi}_i x \in \mathbb{I}(P)$, $\hat{\xi}_i x \in \mathbb{I}(P)$. Now from the previous lemma we can assume $x = \phi_i^{(N)} x_N$ for some x_N as in that lemma. Then $\hat{\phi}_i x = \hat{\phi}_i^{(N+1)} x_N \in \mathbb{I}(P)$ and $\hat{\xi}_i x = \hat{\phi}_i^{(N-1)} x_N \in \mathbb{I}(P)$.

For any $N \geq 0$, we denote by $T_N(P)$ the set of $x \in P$ such that $x = \phi_i^{(N)} x'$ for some $x' \in P(i) \cap {}_\lambda P$ with $(x', x') \in 1 + v^{-1} A$. From the definition we get

$$(a) \quad \hat{\phi}_i(T_N(P)) = T_{N+1}(P)$$

$$(b) \quad \hat{\xi}_i(T_N(P)) = T_{N-1}(P) \text{ for } N \geq 1 \text{ and } \hat{\xi}_i(T_0(P)) = 0.$$

(c) if $N \geq 0$ then $\hat{\phi}_i: T_N \rightarrow T_{N+1}(P)$ and $\hat{\varepsilon}_i: T_{N+1}(P) \rightarrow T_N(P)$ are inverse bijections.

Lemma 8. We have

$$\pm B + v^{-1} \mathbb{I}(P) = \bigsqcup_{N \geq 0} T_N(P) + v^{-1} \mathbb{I}(P).$$

Moreover, the sets $B_N := B \cap (T_N(P) + v^{-1} \mathbb{I}(P))$, $N \geq 0$ form a partition of B .

Proof. By using Lemma 4, Lemma 5, we have $T_N(P) \subseteq \pm B + v^{-1} \mathbb{I}(P)$. Conversely, let $x \in \pm B$.

Then we have $(x, x) \in 1 + v^{-1}A$. Hence we have $x = y' + y''$ where $y'' \in v^{-1} \mathbb{I}(P)$ and $y' = \phi_i^{(N)} x'$ for some $N \geq 0$. Thus $x \in T_N(P) + v^{-1} \mathbb{I}(P)$ and the first assertion follows. To prove the second assertion it suffices to show that $T_{N_1}(P) \cap (T_{N_2}(P) + v^{-1} \mathbb{I}(P)) = \emptyset$ if $N_1 \neq N_2$. Assume that

$\phi_i^{(N_1)} x_1 = \phi_i^{(N_2)} x_2 + v^{-1} z$ where $z \in \mathbb{I}(P)$ and $x_1, x_2 \in P(\omega) \cap {}_A P$ satisfy $(x_1, x_1) \in 1 + v^{-1}A$, $(x_2, x_2) \in 1 + v^{-1}A$.

By Lemma 6, we can write $z = \sum_{N \geq 0} \phi_i^{(N)} z_N$ where $z_N \in \mathbb{I}(P) \cap P(\omega)$. Then

$$\phi_i^{(N_1)} x_1 = \phi_i^{(N_2)} x_2 + v^{-1} \sum_{N \geq 0} \phi_i^{(N)} z_N$$

By Lemma 3 (c), this implies $z_N = 0$ for $N \neq N_1, N_2$, $v^{-1} z_{N_1} = x_1$ and $v^{-1} z_{N_2} = -x_2$. From the last equality we deduce that $(x_2, x_1) = v^{-2} (z_{N_2}, z_{N_1}) \in v^{-2}A$, a contradiction.

Using the previous lemma, we deduce the following:

$$(a) \quad \hat{\phi}_i(\pm B_N + v^{-1} \mathbb{I}(P)) = \pm B_{N+1} + v^{-1} \mathbb{I}(P)$$

$$(b) \quad \hat{\varepsilon}_i(\pm B_N + v^{-1} \mathbb{I}(P)) = \pm B_{N-1} + v^{-1} \mathbb{I}(P)$$

$$(c) \quad \text{if } N \geq 0, \text{ then } \hat{\phi}_i: \pm B_N + v^{-1} \mathbb{I}(P) \rightarrow \pm B_{N+1} + v^{-1} \mathbb{I}(P) \text{ and}$$

$$\hat{\varepsilon}_i: \pm B_{N+1} + v^{-1} \mathbb{I}(P) \rightarrow \pm B_N + v^{-1} \mathbb{I}(P) \text{ are inverse bijections.}$$

Adapted bases

Let P be as in previous section. A basis B of P is called adapted if there exists a partition

$B = \bigsqcup_{n \geq 0} B(n)$ and bijections $\pi_n: B(\omega) \rightarrow B(n)$ for all $n \geq 0$ such that

$$(a) \quad \text{for all } N \geq 0, \quad \bigsqcup_{n \geq N} B(n) \text{ is a basis of } \phi_i^{(N)} P;$$

(b) for any $b \in B(0)$ and any $N \geq 0$, we have $\phi_i^{(N)} b - \pi_N(b) \in \phi_i^{(N+1)} P$.

We assume that the basis B of P is integral, almost orthonormal, and adapted.

Recall the notation $B_N := B \cap (T_N(P) + v^\perp \mathbb{L}(P))$ in the previous section. We know that $B = \bigcup_{N \geq 0} B_N$ is also a partition. Later we will show that $B_N = B(N)$.

Lemma 9. Let $b \in B$. Then $b \in B_0$ if and only if $b \notin \phi_i P$. In particular, $B_0 = B(0)$.

Proof. Assume first that $b \in B_N$ for some $N > 0$. Then $b = \phi_i^{(N)} x' + v^\perp z$ where $z \in \mathbb{L}(P)$ and

$x' \in P$. Since B is adapted, we can write $\phi_i^{(N)} x' = \sum c_b b'$ where b' runs over $B \cap \phi_i^{(N)} P$ and $c_b \in \mathbb{Q}(v)$. We can also write $z = \sum d_{b''} b''$ where b'' runs over B and $d_{b''} \in \mathbb{Z}[v^{-1}]$.

If $b \notin \phi_i^{(N)} P$, then by comparing the coefficients of b we obtain $1 = v^\perp d_b$, a contradiction.

Thus we have $b \in \phi_i^{(N)} P$. Since $N > 0$, we have $b \in \phi_i P$.

Conversely, assume that $b \in \phi_i P$ and $b \in B_0$. Then $b = x' + v^\perp z$ where $z \in \mathbb{L}(P)$, $x' \in P(0)$, and also

$b \in \bigcup_{N \geq 0} \phi_i^{(N)} P(0)$. Using $(P(0), \phi_i^{(N)} P(0)) = 0$ for $N > 0$, we deduce $(x', b) = 0$, hence $(b, b) = v^\perp (z, b) \in v^\perp \Lambda$

(recall that B is a basis of $\mathbb{L}(P)$), a contradiction.

Now to see that $B_0 = B(0)$, we note that $\bigcup_{n \geq 1} B(n)$ is a basis of $\phi_i P$, therefore we have $b \notin \phi_i P$ iff $b \in B(0)$.

Lemma 10 Let $b \in B_0$, $N \geq 0$ and let b' be the unique element of $\pm B$ such that $\hat{\phi}_i^N(b) = b' \bmod v^\perp \mathbb{L}(P)$.

Then $b' = \pi_N b$.

Proof. We write $b = x + v^\perp z$ where $z \in \mathbb{L}(P)$ and $x \in P(0) \cap \perp P$ satisfies $(x, x) \in 1 + v^\perp \Lambda$.

Using Lemma 6, we write $z = \sum_N z_N$ where $z_N \in \mathbb{L}(P) \cap \phi_i^{(N)} P(0)$ for all N . Replacing x by $x + v^\perp z_0$ and z by $z - z_0$, we see that we may assume that z satisfies

$z \in \phi_i P$. The equalities $\hat{\phi}_i^N b = \hat{\phi}_i^N x + v^\perp \hat{\phi}_i^N z$ and $\phi_i^{(N)} b = \phi_i^{(N)} x + v^\perp \phi_i^{(N)} z$,

together with $\hat{\phi}_i^N z \in \mathbb{L}(P)$ and $\phi_i^{(N)} z \in \phi_i^{(N+1)} P$, imply

$$\hat{\phi}_i^N b \equiv \phi_i^{(N)} b \bmod v^\perp \mathbb{L}(P) + \phi_i^{(N+1)} P.$$

By assumption we have $\hat{\phi}_i^N(b) \equiv b' \bmod v^\perp \mathbb{L}(P)$, hence

$$b' \equiv \phi_i^{(N)} b \pmod{v^{-1}\mathbb{Z}(p) + \phi_i^{(N+1)} p}$$

Moreover, we have (as B is adapted)

$$\phi_i^{(N)} b = b_1 \pmod{\phi_i^{(N+1)} p}$$

where $b_1 = \pi_N b \in B$.

We must show that $b' = b_1$. We have $b_1 + c_1 = b' + c$ where $c_1 \in \phi_i^{(N+1)} p$ and $c \in v^{-1}\mathbb{Z}(p)$.

Note that $b_1 \notin \phi_i^{(N+1)} p$, since otherwise $\phi_i^{(N)} b \in \phi_i^{(N+1)} p$, so $b \in \phi_i p$, contradicting to Lemma 9.

Hence, if we express $b_1 + c_1$ as a $\mathbb{Q}(v)$ -linear combination of elements of B , the element $b_1 \in B$

will appear with coefficient 1. On the other hand, if we express $b' + c$ as a $\mathbb{Q}(v)$ -linear combination of elements of B , the coefficients should lie in $v^{-1}\mathbb{Z}(v)$ except that of $\pm b'$.

This forces $b_1 = \pm b'$. If $b_1 = -b'$ then we get $2b_1 + c_1 = c'$ and $\pm b_1$ appears in the LHS with coefficient 2, and in the RHS with coefficient in $v^{-1}\mathbb{A}$, a contradiction. Hence $b_1 = b'$ and the lemma follows.

We now show that the action of the operators $\hat{\phi}_i, \hat{\varepsilon}_i$ on B is described up to elements in $v^{-1}\mathbb{Z}(p)$ in terms of the bijections π_n .

Prop 11 Let $b \in B(N)$, and $b_0 \in B(0)$ be the unique element such that $\pi_N b_0 = b$. Then we have

$$\hat{\phi}_i b \equiv \pi_{N+1} b_0 \pmod{v^{-1}\mathbb{Z}(p)} \quad \text{and} \quad \hat{\varepsilon}_i(b) \equiv \pi_{N+1} b_0 \pmod{v^{-1}\mathbb{Z}(p)} \text{ if } N \geq 1, \quad \hat{\varepsilon}_i(b) \equiv 0 \pmod{v^{-1}\mathbb{Z}(p)} \text{ if } N=0.$$

In particular, we have $B_N = B(N)$ for all N .

Proof. This follows from Lemma 10.

Operators $\hat{\phi}_i, \hat{\varepsilon}_i: f \rightarrow f$

Recall that we can regard f as a $\mathcal{U}$ -module. Thus for each $i \in \mathbb{I}$, f with the operator

$\phi_i, \varepsilon_i: f \rightarrow f$ is an object in $\mathcal{D}_i$. Hence $\hat{\varepsilon}_i, \hat{\phi}_i: f \rightarrow f$ are well-defined.

Note that the form $(\cdot, \cdot)$ on f is admissible.

For a fixed $i \in \mathbb{I}$, we define a $\mathbb{Q}(v)$ -basis B^i of f as follows. By definition $B^i = \bigcup_{t \geq 0} B^i(t)$

where $B^{i(0)}$ is any subset of $B_{i;0}$ such that $B_{i;0} = B^{i(0)} \sqcup B^{i(0)}$, and $B^{i(t)}$ is the image of $B^{i(0)}$ under $\pi_{i,t}: B_{i;0} \xrightarrow{\sim} B_{i;t}$. Then we have $B = B^i \sqcup B^i$ and B^i is adapted. By definition we see that B^i is almost orthonormal and generate Δ^f .

Let $\mathbb{L}(f) = \{x \in \Delta^f \mid (x, x) \in A\}$. Then $\mathbb{L}(f)$ is the $\mathbb{Q}(\Gamma^+)$ -submodule generated by Θ^i .

Lemma 12. (a) Δ^f is stable under the operators $\epsilon_i, \phi_i^{(t)}: f \rightarrow f$ for any $i \in \mathbb{I}$.

(b) $\mathbb{L}(f)$ is stable under the operators $\hat{\phi}_i, \hat{\epsilon}_i: f \rightarrow f$ for any $i \in \mathbb{I}$.

Prop. 13. Let $b \in B^{i(t)}$, and $b_0 \in B^{i(0)}$ be the unique element such that $\pi_{i,t} b_0 = b$.

Then we have $\hat{\phi}_i(b) \equiv \pi_{i,t+1}(b_0) \bmod \mathfrak{v}^1 \mathbb{L}(f)$, and $\hat{\epsilon}_i(b) \equiv \pi_{i,t+1}(b_0) \bmod \mathfrak{v}^1 \mathbb{L}(f)$ if

$t \geq 1$, $\hat{\epsilon}_i(b) \equiv 0 \bmod \mathfrak{v}^1 \mathbb{L}(f)$ if $t = 0$.